



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani

**Pilani Campus
AUGS/ AGSR Division**

FIRST SEMESTER 2026-2027 COURSE HANDOUT

Date: 01/08/2026

In addition to part I (the general handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course no. : BITS F462/BITS U462
Course title : Renewable energy
Instructor in charge : MANOJ KUMAR SONI

Course Description:

Introduction of renewable energy, advantages, potential, status of development, broad details of different renewable energy systems such as solar, wind, biomass, microhydel, geothermal etc; Renewable energy development policy, Renewable energy industries, international co-operation, HRD and career growth opportunities, consultancy areas and future thrust areas in renewable energy development.

Scope:

State-of-the-art treatment of different sources of renewable energy, their utility, and techno-economic assessment.

Course Objective:

1. Evaluation: Evaluate: To determine the power output and evaluate the performance of simple thermodynamic cycles for power generation technologies in terms of thermal efficiency.
2. Evaluation: Evaluate: To be able to analyze and evaluate the performance of coal-based or gas-based power generation technologies.
3. Evaluation: Estimate: To be able to estimate the performance of wind, geothermal, ocean, and solar power technologies with/without energy storage for a given geographical site, in terms of their efficiency or potential.
4. Analysis: Analyze: To be able to analyze bio-chemical and thermo-chemical conversion technologies of biomass and organic waste in terms of their potential and conversion efficiency.
5. Analysis: Analyze: To analyze and compare the performance of fuel cell technologies in terms of operating conditions, efficiency and applications.

Text book:

1. Kothari, D.P., Singal, K. C., Ranjan, Rakesh, *Renewable Energy Sources and Emerging Technologies*, Third Edition, PHI Learning Private Ltd., New Delhi, 2022. ISBN: 978-81-203-4470-9

Reference books:

1. G. N. Tiwari, and M. K. Ghosal, *Renewable energy resources: basic principles and applications*. Narosa Publishing House, 2005.



Course lecture plan:

Lecture No.	Learning objectives	Topics to be covered	Chapter Textbook
1-2	Importance and applicability of renewable energy sources.	Energy Resources and Their Utilization, Environmental Aspects of Electric Energy Generation.	1, 2
3-11	Technology and economics of solar thermal systems.	Solar Radiation and Its Measurement, Solar Thermal Energy Collectors, Solar Thermal Energy Conversion Systems.	3, 4, 5
12-13	Technology and economics of photovoltaic systems.	Solar Photovoltaic System	6
14-17	Technology and economics of wind power.	Types of turbines and their aerodynamics. Theories of wind turbines. WECS components and state of art Feasibility of large wind turbines. Wind Farms.	7, 8
18-21	Technology and economics of small hydro.	Types and theories of small, mini, and micro-hydro turbines. Hydroelectric systems and advantages at small hydro levels. Conversion apparatus and feasibility.	9
22-25	Technology and economics of geothermal, wave and tidal energy systems	Geothermal Energy, Wave, and tidal turbines. Harnessing wave energy, applicability, and feasibility, Electric Power Generation by Ocean Energy.	10, 11
26-29	Technology and economics of biogas and bio power	Biofuel production and gasification. Biopower generation methods with technical details: steam, combined cycle, cogeneration.	12
30-33	Fuel Cells, Hydrogen and Hybrid systems	Fuel Cells, Hydrogen Energy System, Hybrid Systems.	13, 14, 15
34	Environmental Impact	Environmental impacts of renewable energy sources and remedial measures. Global Climate Change	16
35-37	Cost analysis techniques.	Cost calculations based on external costs, payback periods, cost/benefit analysis. Issues in multi-unit planning.	12
38	Case studies		



Evaluation scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of Component
Mid Semester Exam	90	25	TBA	Closed Book
Lecture Test (Surprise)*	10	10	During class hours	Closed Book
Activity-based evaluation/ case studies/ Assignments/ projects		25	Announced later	
Comprehensive Exam	180	40	TBA	Closed and Open Book

*During the scheduled class hour

Reading assignments:

From time to time, reading assignments will be given to the students. These reading assignments are part of the course, and questions may appear in tests/examinations in these portions, too.

Chamber consultation hours: To be announced in the class.

Course notices: To be displayed Nalanda as and when appropriate.

Makeup Policy: Makeup will be given only to genuine cases. The request application should reach the Instructor-in-charge before the commencement of the scheduled test.

Minimum passing Marks criteria: *The passing marks for this course will be determined based on one of the two criteria: 20% of the median of the post-comprehensive examination score OR 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course. NC may also be awarded if a student misses regular evaluation components (except lecture surprise tests) even if he/she satisfies the above criteria.*

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Renewable energy**